

ASSIGNMENT 5

Assemble and Disassemble a PC**Practical**

Q1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

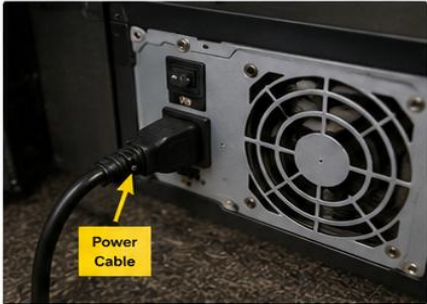
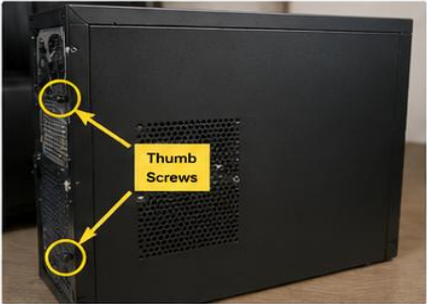
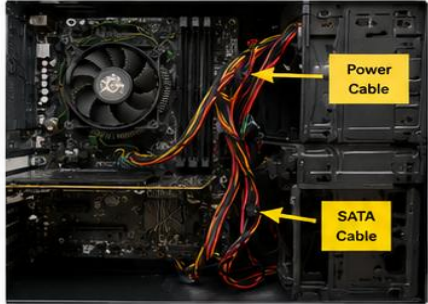
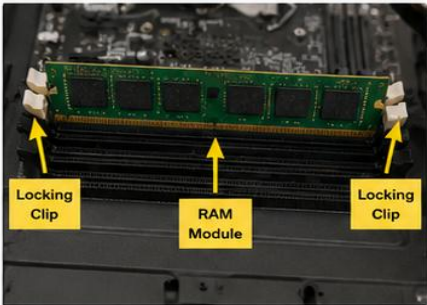
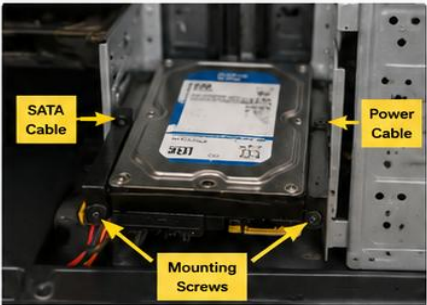
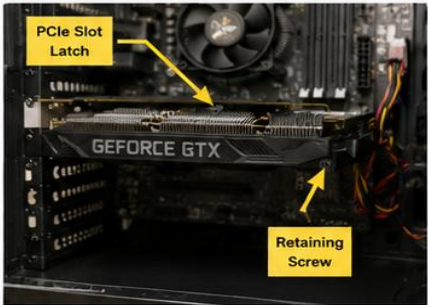
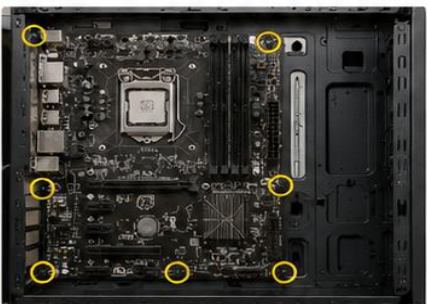
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Component	Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The brain of the computer that processes instructions and performs calculations.
CPU Cooler (Fan/Heatsink)	Keeps the processor cool by removing excess heat during operation.
RAM (Memory)	Temporarily stores data and programs currently being used by the CPU.
Graphics Card (GPU)	Processes graphics and displays images, videos, and games on the monitor.
Storage Drive (HDD/SSD)	Stores the operating system, software, and personal files permanently.
Power Supply Unit (PSU)	Converts AC power into DC power and supplies electricity to all components.
Case Fans	Improve airflow by bringing cool air in and removing hot air from the cabinet.
Optical Drive (Optional)	Reads and writes CDs, DVDs, or Blu-ray discs.
Expansion Cards	Add extra features such as Wi-Fi, sound, or additional USB ports.
SATA/Data Cables	Connect storage devices to the motherboard for data transfer.
Power Cables	Deliver power from the PSU to the motherboard, storage drives, and other hardware.

Q2. Take clear photos or draw labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

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Desktop PC Disassembly – Step by Step		
Step 1: Unplug the Power A close-up photo of the back of a PC case showing the power cable being unplugged from the power supply. A yellow label 'Power Cable' points to the cable. Turn off the computer and unplug the power cable. Disconnect all other external cables.	Step 2: Remove the Side Panel A photo of the back of a PC case with the side panel removed. Two yellow circles highlight the thumb screws. A yellow label 'Thumb Screws' points to them. Remove the thumb screws from the back of the cabinet and slide the side panel backward to take it off.	Step 3: Disconnect Cables A photo of the interior of a PC case with the side panel removed. Yellow labels 'Power Cable' and 'SATA Cable' point to the respective cables being disconnected. Disconnect power cables from the motherboard, GPU and drives. Remove SATA data cables from HDD/SSD.
Step 4: Remove the RAM A photo of a RAM module being removed from its slot. Yellow labels 'Locking Clip', 'RAM Module', and 'Locking Clip' point to the respective parts. Press the locking clips on both sides of the RAM slot. The module will pop up. Gently lift it out.	Step 5: Remove HDD/SSD A photo of a 3.5-inch HDD being removed from its bay. Yellow labels 'SATA Cable', 'Power Cable', and 'Mounting Screws' point to the respective parts. Unscrew the drive from the drive bay. Disconnect the SATA data cable and power cable. Remove the drive.	Step 6: Remove Graphics Card (GPU) A photo of a GeForce GTX graphics card being removed from its slot. Yellow labels 'PCIe Slot Latch' and 'Retaining Screw' point to the respective parts. Remove the retaining screw at the back. Press the PCIe slot latch and gently pull the graphics card out.
Step 7: Remove CPU Cooler A photo of a CPU cooler being removed from the motherboard. Yellow labels 'Fan Cable' and 'Mounting Screws' point to the respective parts. Unplug the CPU fan cable. Loosen the mounting screws (or push the clips) and carefully lift the cooler off the CPU.	Step 8: Remove Motherboard A photo of the motherboard being removed from the case. Yellow circles highlight the mounting screws. Disconnect all remaining cables from the motherboard. Remove the mounting screws and carefully lift the motherboard out of the cabinet.	Step 9: Organize Components A photo of all the disassembled components (motherboard, RAM, HDD, GPU, cooler, etc.) laid out on a surface. A yellow label 'Keep all components and screws in order. Store screws in a small box so nothing gets lost.' points to the components. Keep all components and screws in order. Store screws in a small box so nothing gets lost.

- Step 1: Unplug the Power
- Turn off the computer and unplug the power cable.
 - Disconnect all external devices such as the monitor, keyboard, mouse, and network cable.
- Step 2: Remove the Side Panel
- Unscrew the side panel of the PC cabinet.
 - Slide the panel backward and remove it carefully to access the internal components.

➤ Step 3: Disconnect Cables

- Disconnect the 24-pin motherboard power cable, CPU power cable, SATA cables, and GPU power cable.
- Keep the cables organized for easy reassembly.

➤ Step 4: Remove the RAM

- Press the locking clips on both sides of the RAM slot.
- Gently lift the RAM module out of the motherboard.

➤ Step 5: Remove the HDD/SSD

- Unscrew the HDD/SSD from its mounting bracket.
- Disconnect the SATA data cable and power cable, then remove the drive.

➤ Step 6: Remove the Graphics Card (GPU)

- Unscrew the graphics card from the case.
- Press the PCIe slot latch and carefully pull the GPU out.

➤ Step 7: Remove the CPU Cooler

- Disconnect the CPU fan cable.
- Unscrew or unclip the cooler and carefully lift it off the CPU.

➤ Step 8: Remove the Motherboard

- Disconnect any remaining cables.
- Remove the motherboard mounting screws and carefully lift the motherboard out of the cabinet.

➤ Step 9: Organize the Components

- Arrange all removed components neatly.
- Store screws in a small container to prevent them from getting lost.

Q3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

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❖ Desktop PC Reassembly Steps

➤ Step 1: Install the Motherboard

- Place the motherboard inside the cabinet and secure it with screws.
- This is done first because all other components connect to the motherboard.

➤ Step 2: Install the CPU Cooler

- Mount the CPU cooler on the processor and connect the CPU fan cable.
- This keeps the CPU cool and prevents overheating during operation.

➤ Step 3: Install the RAM

- Insert the RAM module into the correct DIMM slot until the locking clips click.
- RAM must be installed before the system can boot properly.

➤ Step 4: Install the HDD/SSD

- Mount the storage drive in its bay and connect the SATA data cable and power cable.
- The storage drive contains the operating system and files needed to start the computer.

➤ Step 5: Install the Graphics Card (GPU)

- Insert the GPU into the PCIe x16 slot and secure it with a screw.
- Connect the PCIe power cable if required for proper operation.

➤ **Step 6: Connect Power Cables**

- **Connect the 24-pin ATX motherboard power cable and the 8-pin CPU power cable.**
- **These cables supply power to the motherboard and processor.**

➤ **Step 7: Connect Front Panel and Other Cables**

- **Connect the power switch, reset switch, LED connectors, USB headers, and audio cables.**
- **These connections allow the front buttons and ports to function correctly.**

➤ **Step 8: Close the Cabinet**

- **Reattach the side panel and tighten the screws.**
- **This protects the internal components and improves airflow.**

➤ **Step 9: Power On and Test**

- **Connect the power cable and switch on the PC.**
- **Verify that the system boots successfully and all components are detected, confirming the reassembly was completed correctly.**

Q4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

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❖ **PC Assembly/Disassembly Safety Checklist:**

- 1. Turn Off the Computer:** Prevents electrical damage and protects the user from electric shock.
- 2. Unplug the Power Cable:** Ensures that no electricity is supplied to the computer during work.

- 3. Wear an Anti-Static Wrist Strap:** Prevents electrostatic discharge (ESD) that can damage computer components.
 - 4. Work on a Clean, Flat Surface:** Reduces the risk of dropping or damaging hardware and losing small parts.
 - 5. Use a Phillips Screwdriver:** Used to safely remove and install screws on the PC cabinet and components.
 - 6. Keep Screws in a Small Container:** Prevents screws from being lost and makes reassembly easier.
 - 7. Handle Components by the Edges:** Avoids touching sensitive chips and connectors, reducing the risk of damage.
 - 8. Disconnect All External Devices:** Prevents cable damage and allows easy access to the computer.
 - 9. Avoid Liquids Near the PC:** Protects electronic components from short circuits and damage.
 - 10. Check All Connections Before Powering On:** Ensures every component is installed correctly and helps prevent boot problems or hardware damage.
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